Computer Architectures
Exam of 01/07/2025

A2

Question 1 (6 points)

A linear congruential generator (LCG) is an algorithm that produces a sequence of pseudo-random numbers. Write the `nextElementLCG` subroutine, in ARM assembly, that returns the *n*-th element of a sequence, computed as follows:

$$X_n = ((aX_{n-1} + c) \text{ EXOR } n) \bmod m$$

The subroutine receives in input (in the specified order):

1. the previous element of the sequence X_{n-1}
2. the multiplier a
3. the increment c
4. the mask n
5. the module m

In order to test the procedure, define an array of bytes with *DIM* elements in a readwrite area of memory (*DIM* is a constant that can be freely chosen). Then, write a loop of *DIM* iterations in the `Reset_Handler` to fill the array. In each iteration of the loop, you call the `nextElementLCG` subroutine with the appropriate parameters.

Example. With the seed 1, a = 131, c = 7, n = index of the loop, m = 255, the first 10 elements are:

iteration n	$aX_{n-1} + c$	$(aX_{n-1}$ + c) EXOR n	$X_n = ((aX_{n-1} + c)$ EXOR n) mod m	Hex
0	131 * 1 + 7 = 138	138 EXOR 0 = 138	138 mod 255 = 138	8A
1	131 * 138 + 7 = 18085	18085 EXOR 1 = 18084	18084 mod 255 = 234	EA
2	30661	30663	63	3F
3	8260	8263	103	67
4	13500	13496	236	EC
5	30923	30926	71	47
6	9308	9306	126	7E
7	16513	16518	198	C6
8	25945	25937	182	B6
9	23849	23840	125	7D

Question 2 (6 points)

Extend the project to control leds 8-11 based on the values generated by the LCG.
First, start timer 0 in order to generate an interrupt every 3 seconds.
Then, in the handler of timer 0, call the `nextElementLCG` subroutine to obtain the *n*-th element of the sequence. Since the value ranges in [0, 254], compute the modulo 4 of such value:

- if value mod 4 == 0, switch on led 11
- if value mod 4 == 1, switch on led 10
- if value mod 4 == 2, switch on led 9
- if value mod 4 == 3, switch on led 8

Use the same parameters as in the example: seed 1, a = 131, c = 7, n = index of the loop, m = 255.
The `nextElementLCG` subroutine is called only 10 times (i.e., during the first 10 interrupts of timer 0). The leds are switched on in the following order:

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iteration ***n***	X_n	X_n **mod 4**	**Led on**
0	138	2	9
1	234	2	9
2	63	3	8
3	103	3	8
4	236	0	11
5	71	3	8
6	126	2	9
7	198	2	9
8	182	2	9
9	125	1	10

Important: Ensure that only one led is on at any given time. Before switching on a new led, turn off the previously lit led (from the previous timer interrupt).

Question 3 (6 points)

Extend the project to implement a rhythm game using the joystick on the board. The game will involve moving the joystick in response to the leds being switched on.
When a led lights up, the player must move the joystick accordingly:

- led 11: move the joystick up
- led 10: move the joystick left
- led 9: move the joystick right
- led 8: move the joystick down

Only the first movement of the joystick has to be considered (further movements within the 3-second interval are ignored). If the movement corresponds to the correct direction for the lit led, increment the variable `num_correct`, otherwise increment `num_wrong`. Regardless of the correctness, after the joystick movement the led is immediately switched off.
The game continues for 10 sequence elements. After 10 elements, the game ends and the player's score is evaluated. If `num_correct` > `num_wrong`, the player wins: turn on led 4 to indicate victory. Otherwise, the player looses: turn on led 5 to indicate defeat.